Lab #1

Getting Started with Zynq-7000 SoC development platforms

Sept. 28, 2026

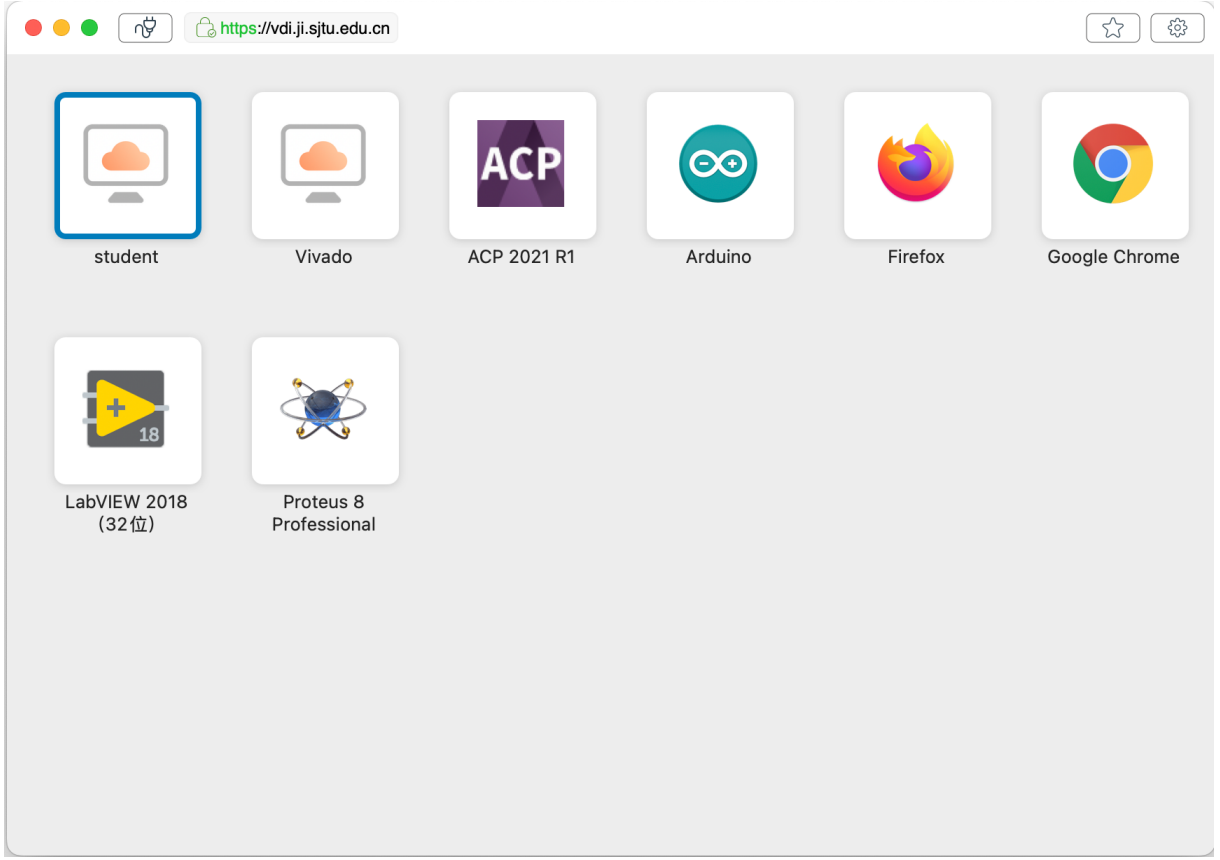
Group Assignment

Group 1 Xia Tian 夏天 Yu Haibo 俞海博	**Group 6** Wu Junying 吴俊莹 Liao Yicheng 廖逸澄	**Group 11** Wang Sicheng 汪思成 Tong Siqi 童思齐
Group 2 Zhao Tong 赵同 Yuan Haotong 袁昊桐	**Group 7** Duan Yikun 段熠坤 Zhao Ziyue 赵子越	**Group 12** Xiao Jiakai 肖嘉恺 Suo Junkai 索浚凯
Group 3 Liang Shirui 梁诗睿 Yao Shuya 姚舒雅	**Group 8** DAYANA KHEGAY ANASTASIIA IANKE	**Group 13** HALFDAN ELVIS FRIDAL-OMERAGIC DAVID EDVIN WELZIEN
Group 4 Ding Ruohan 丁若菡 Zhu Chiheng 朱持衡	**Group 9** ALEXANDRE BEAUDIN YUSIF MAMMADOV	**Group 14** Song Pengtao 宋鹏涛 Xu Yizhe 徐熠喆
Group 5 Yang Tianzhe 杨天哲 NICODEMUS NOEL LEE	**Group 10** ESKIL TRONENG EVEN MALVIK FJELDSTAD	**Group 15** HONG JAY YAP

Each group will share one FPGA board. Board type will be distributed during the lab session.

Virtual Desktop (Still getting set up)

- You can use it as mentioned in **Lab 0 manual**.
- You may need **SJTU VPN** off campus.
- If you need it but **haven't sign up in feishu form** or **have issues to access it**, please inform me asap.
- Please be aware that there might be **some latency** when connected to the server.



Lab Schedule

- https://oc.sjtu.edu.cn/courses/97854/assignments/syllabus
- Due date to submit lab deliveries and lab session time are **different**.

Week	Date	Lab Contents	Due Date
1	No in-person lab	Lab #0 Getting to know AMD-Xilinx Arty Z7 & Vivado	Fri. 9/25 23:59
2	Mon. 9/28	Lab #1 Getting Started with Arty Z7 SoC Development Platform	Fri. 10/9 23:59
4	Mon. 10/12	Lab #2 Design with High Level Synthesis on Arty Z7 SoC Development Platform (1.5 Weeks)	Fri. 10/23 23:59
5	Mon. 10/19	Lab #2 Design with High Level Synthesis on Arty Z7 SoC Development Platform (1.5 Weeks)	Fri. 10/23 23:59
6	Mon. 10/26	Lab #3 PYNQ Overlays & Optimizing HLS Performance through Pipelining (1.5 Weeks)	Fri. 11/6 23:59
7	Mon. 11/2	Lab #3 PYNQ Overlays & Optimizing HLS Performance through Pipelining (1.5 Weeks)	Fri. 11/6 23:59
8	Mon. 11/9	Lab #4 Getting Started with the Automated ASIC Design Flow based on OpenRoad	Mon. 11/23 23:59
9	Mon. 11/16	Lab #5 Advanced Explorations with the Automated ASIC Design Flow based on OpenROAD	Mon. 11/30 23:59

Logistics

- **Absence notification:**
  - If you can't make it, please justify your reasons and let the instructor/lab management know at least a week ahead!
- **Attendance:**
  - You can leave early if needed, but attendance for lecture is required.
- **Lab deliverables**
  - include but are not limited to reports, codes, demos and peer evaluations.
- **Lab guidelines:**
  - Internet search is allowed, especially for bugs, interesting ideas, but not for direct copy and paste.
  - Please read the documents (references) carefully before you ask questions.
  - Debug should be a fun process instead of being a painful headache.

Lab logistics

- **Lab resources:**
  - Please **take good care of** the board, cables and other accessories. You will not be able to get grades and required to cover the cost of the board if your lab kit is not returned nor pieces are broken/missing.
  - Try to understand why you do it before you do it.
  - Piazza: Use Piazza wisely and properly to discuss questions. Directly post solution codes are not allowed.
- **Teamwork:**
  - Teamwork is very important, sharing what you learned with each other is valuable.
  - Even you are not working on certain part, you should know about it.
  - You need to plan on how you divide the workload across your team.

Zynq-7000 architecture

- 650MHz dual-core Cortex-A9 processor
- DDR3 memory controller with 8 DMA channels and 4 High Performance AXI3 Slave ports
- High-bandwidth peripheral controllers: 1G Ethernet, USB 2.0, SDIO
- Low-bandwidth peripheral controller: SPI, UART, CAN, I2C
- Programmable from JTAG, Quad-SPI flash, and microSD card
- Programmable logic equivalent to Artix-7 FPGA

Zynq-7000 architecture

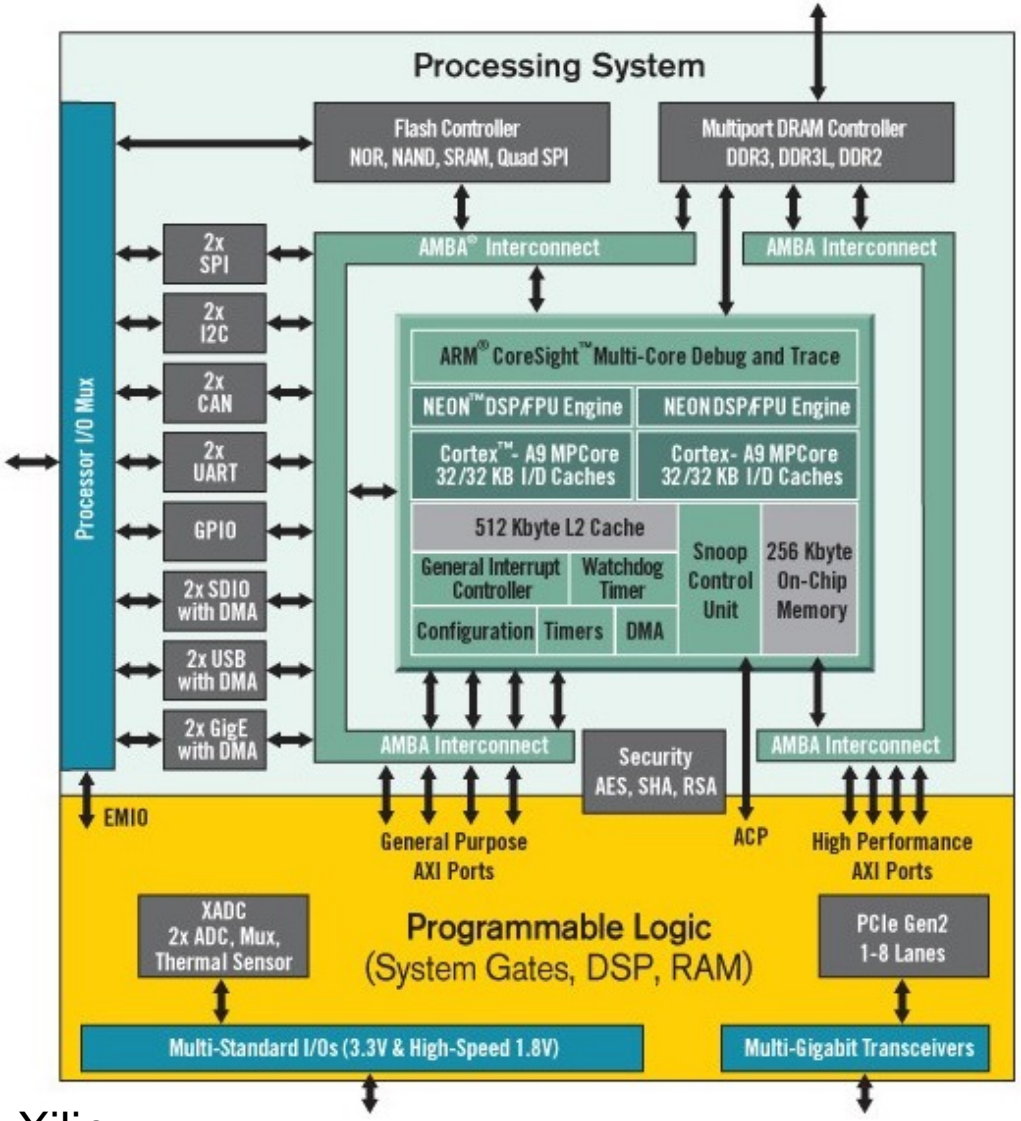


Image: Xilinx

Arty Z7

Product Variant	Arty Z7-10	Arty Z7-20
Zynq Part	XC7Z010-1CLG400C	XC7Z020-1CLG400C
1 MSPS On-chip ADC	Yes	Yes
Look-up Tables (LUTs)	17,600	53,200
Flip-Flops	35,200	106,400
Block RAM	270 KB	630 KB
Clock Management Tiles	2	4
Available Shield I/O	26	49

Arty Z7-20

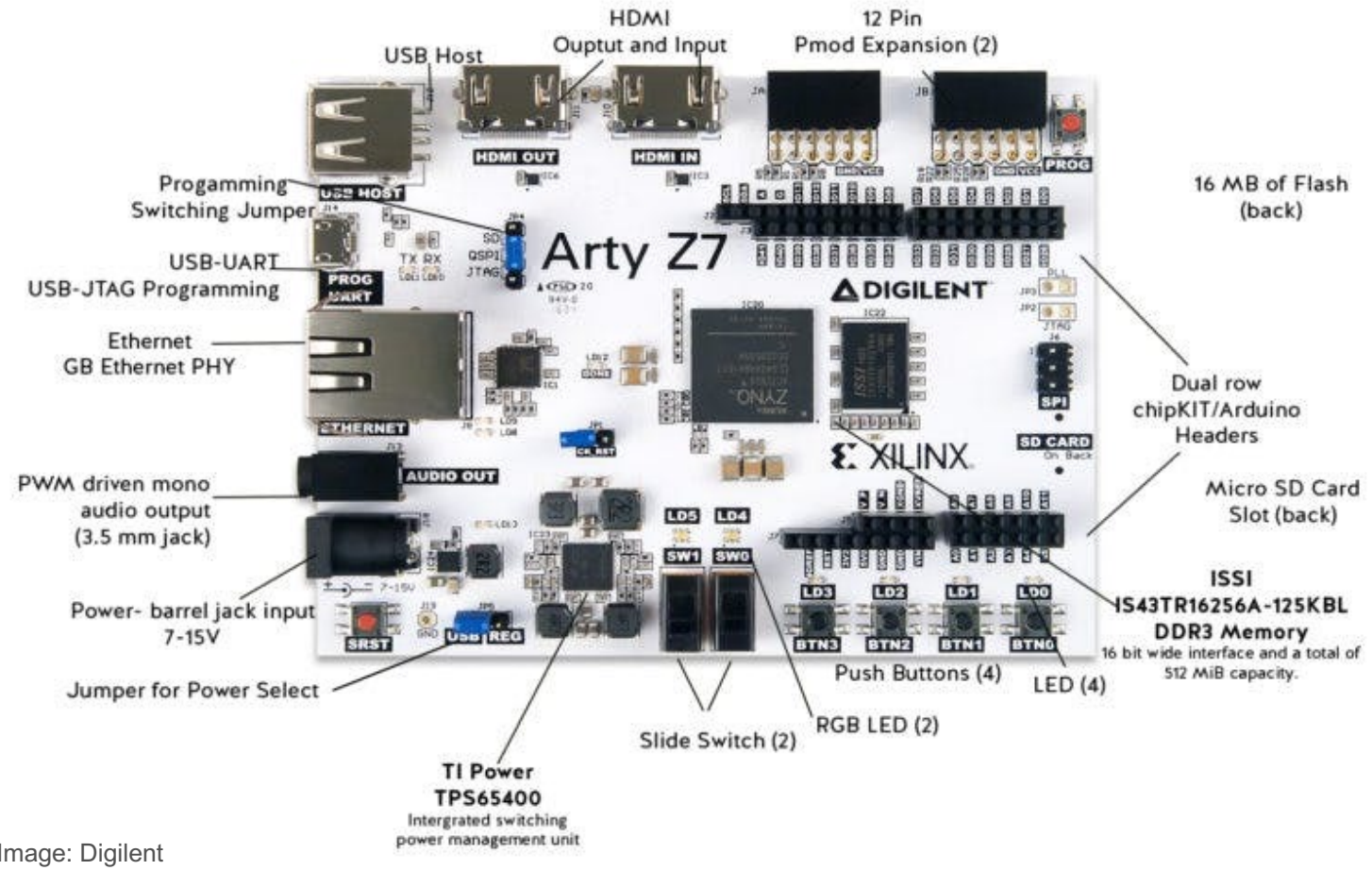


Image: Digilent

PYNQ-Z1

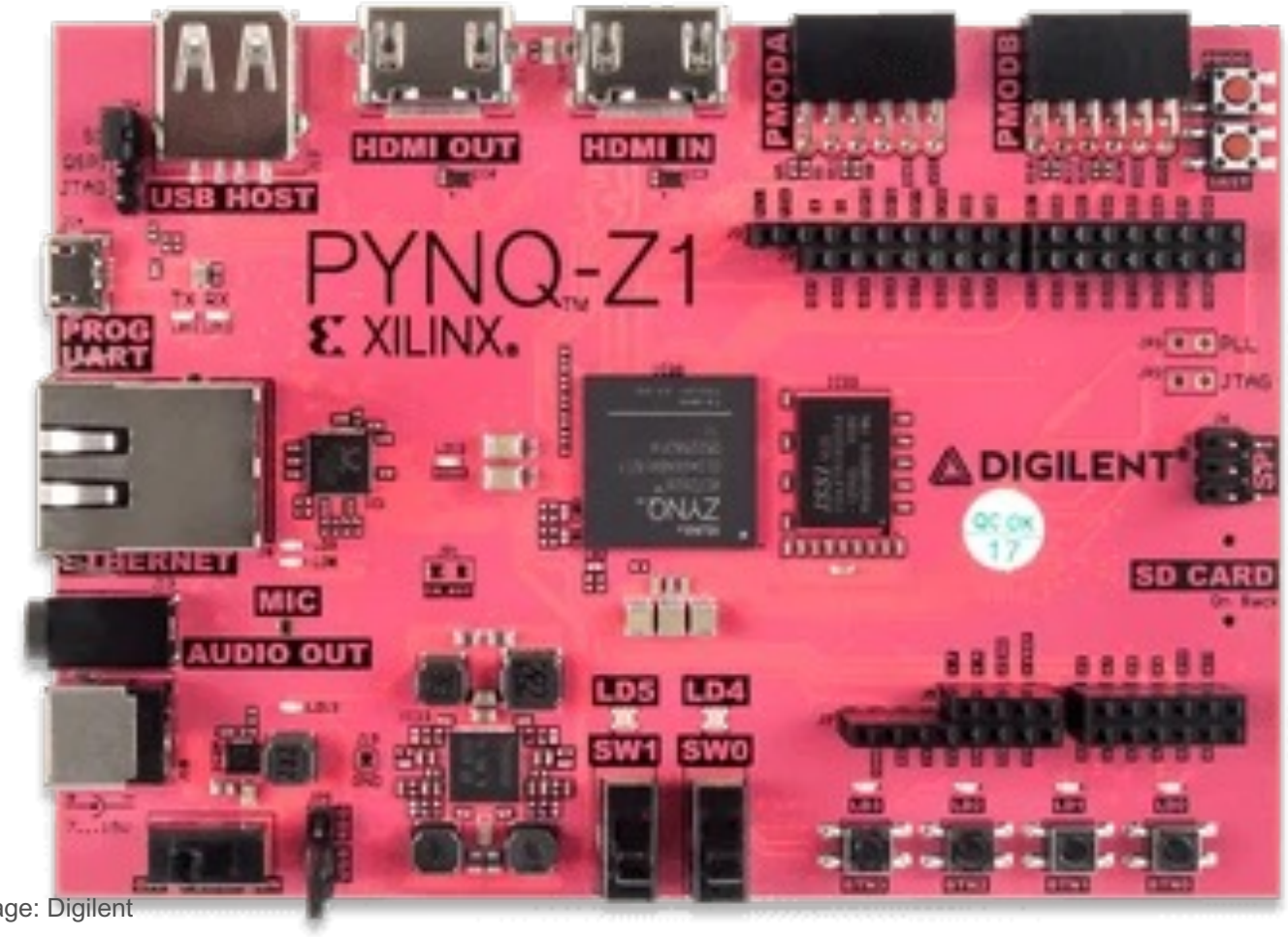


Image: Digilent

Board Resources

Arty Z7-20

Memory

512MB DDR3 with 16-bit bus @ 1050Mbps
16MB Quad-SPI Flash with factory programmed 48-bit EUI-48/64 compatible identifier
microSD slot

USB and Ethernet

Gigabit Ethernet PHY
USB-JTAG programming circuitry
USB-UART bridge
USB OTG PHY, host only

PYNQ-Z1

Memory

512 MB DDR3 with 16-bit bus @ 1050 MT/s
16MB Quad-SPI Flash with factory programmed EUI-48/64 compatible identifier
MicroSD slot

USB and Ethernet

USB-JTAG programming circuitry
USB-UART bridge
USB OTG PHY, host only
Gigabit Ethernet PHY

Sources: Digilent Arty Z7 Reference Manual; Digilent PYNQ-Z1 product page

I/O, Power, and Expansion

Arty Z7-20

Audio and Video

HDMI sink port (input)
HDMI source port (output)
PWM driven mono audio output with 3.5mm jack

Power

Powered from USB or any 7V-15V external power source

Switches, Push-buttons, and LEDs

4 push-buttons
2 slide switches
4 LEDs
2 RGB LEDs

PYNQ-Z1

Audio and Video

Electret microphone with PDM output
3.5mm mono audio output jack, PWM format
HDMI sink port (input)
HDMI source port (output)

Power

Powered from USB or any 7V-15V source

Switches, Push-buttons, and LEDs

4 push-buttons
2 slide switches
4 LEDs
2 RGB LEDs

Expansion Connectors

Two standard Pmod ports
Arduino/chipKIT shield connector

Sources: Digilent Arty Z7 Reference Manual; Digilent PYNQ-Z1 product page

Software Support

- Xilinx Vivado and Vitis are used for Lab 1.
  - Both Arty Z7-20 and PYNQ-Z1 follow the same Vivado/Vitis flow.
  - Use the manual that matches your assigned board.
- PYNQ Linux/Jupyter is a board feature, but it is not used in Lab 1.

Boot Modes

- Both Arty Z7-20 and PYNQ-Z1 support programming from JTAG, Quad-SPI flash, and microSD card. Lab 1 uses JTAG through Vivado/Vitis.

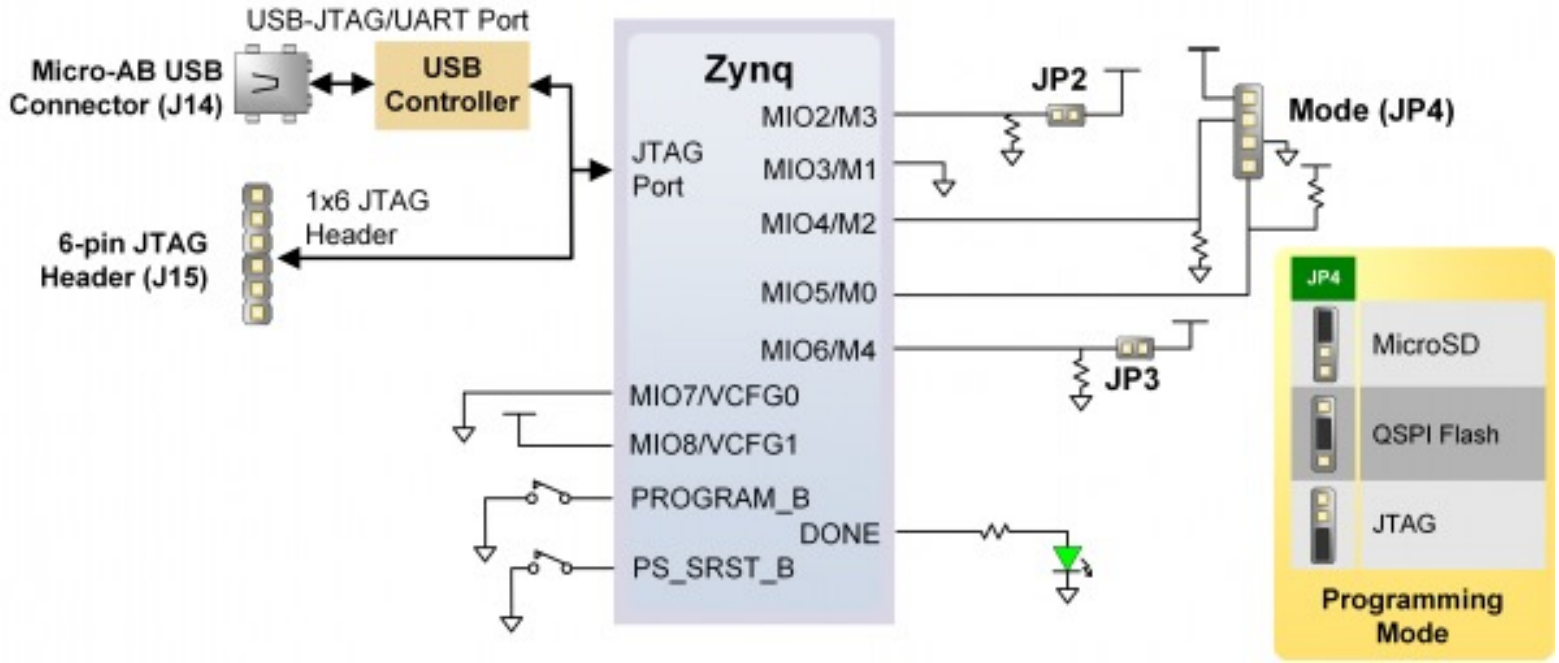
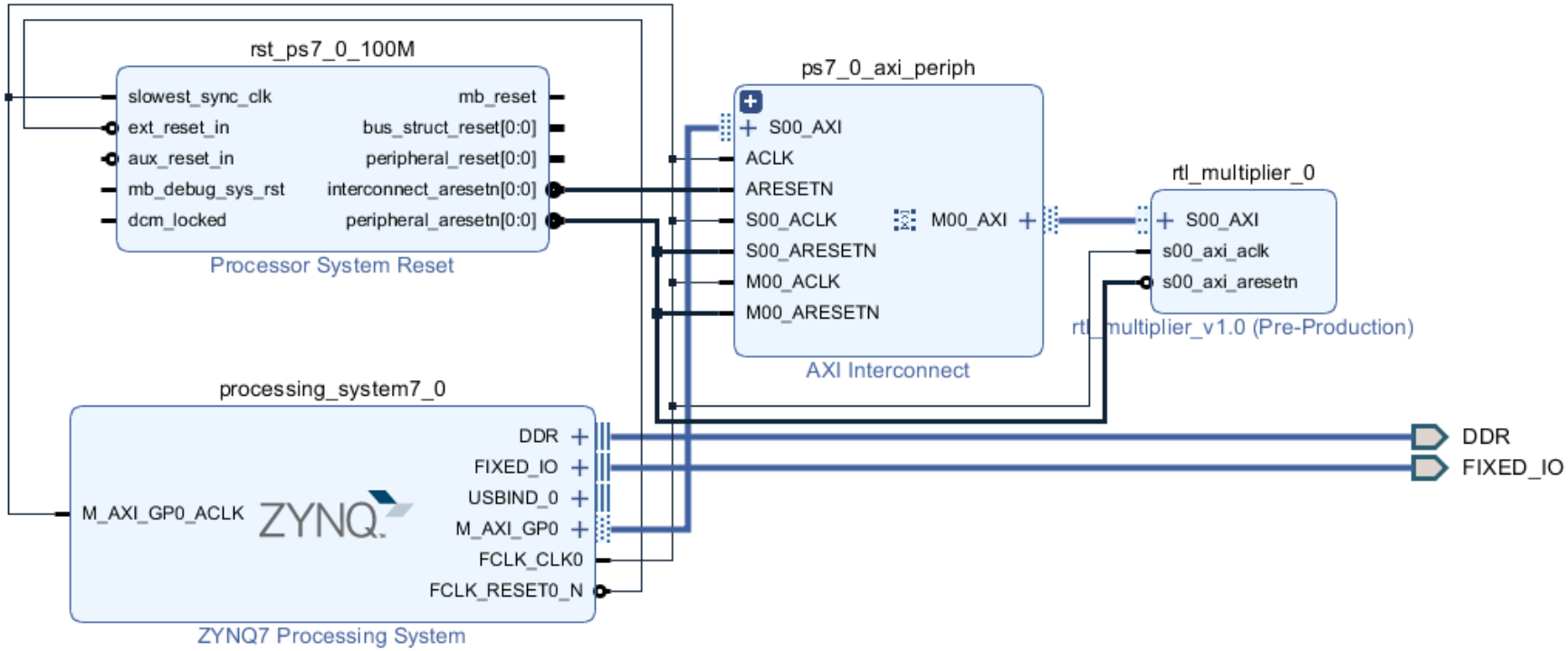


Image: Diligent

AXI Interconnect

The communication between the processing system (PS) and the programmable logic (PL) is handled by the **AXI protocol.**



Tcl

- Tcl (pronounced “tickle”) is a high-level, general-purpose, interpreted, dynamic programming language created in 1988. The popular combination of Tcl with the Tk extension is referred to as Tcl/Tk and enables building a GUI natively in Tcl. Tcl/Tk is included in the standard Python installation in the form of Tkinter.
- **Widely used in Electronic Design Automation (EDA)** for scripting and automating tasks within EDA tools such as Xilinx Vivado
- You will use Tcl in Vivado 2022.1 Tcl Shell most of the time.
- See: *Reduced Tcl Tutorial All-in-One.pdf* on Canvas

```
start_gui
create_project lab1_1 F:/Documents/GitHub/ECE4810J_FA2022_TA/Lab1/lab1_1
 ↪  -part xc7z020clg400-1
set_property board_part digilentinc.com:arty-z7-20:part0:1.1
 ↪  [current_project]
```

Lab #1 Overview

Goals

- Get familiar with the board
- Set up the environment correctly
- Be able to build a basic Zynq system on the board
- Be able to create a custom IP in Verilog
- Be able to use AXI bus to connect an IP block with the Zynq programing system

Lab #1 Overview

- The overalll design flow:
  - Vivado(hardware design)
    - Create a block design
    - Add the Zynq PS & pre-built Ips
    - Add your own custom IP
    - Connect everything
    - Generate a Bitstream(.bit) file
  - Export hardware:
    - Package the entire hardware deisgn into a single .xsa file
  - Vitis
    - Import the .xsa file
    - Write C code to control the processor and communicate with your hardware in the PL
    - Build, download, and run on the assigned board.

Deliverables

- Details are in the Lab #1 description
- Peer evaluation form (individual grade)

References

- https://digilent.com/reference/programmable-logic/arty-z7/reference-manual
- https://digilent.com/shop/pynq-z1-python-productivity-for-zynq-7000-arm-fpga-soc/
- Yichen Cai, 2023 Lab #1
- Fan Hu, 2024 Lab #1